Alireza Noroozi

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Education

Koç University October 2025 – June 2027
MSc in Computer Engineering *GPA: 3.48/4.0*

Sharif University September 2020 – June 2024
Bachelor of Science in Computer Engineering *GPA: 17.32/20 (3.46/4.0)*

Publications

- **A. Noroozi**, Ö. Keskin, A. Gürsoy, Y. Karami, and H. Khakzad, IDPepDesign: Insertion-Deletion is All Peptide Design Needs, submitted to NeurIPS 2026.
- **A. Noroozi** and A. Gürsoy, AMP-DiT: Antimicrobial Peptide Design with Denoising Diffusion Transformers and Classifier Guidance, GenBio Workshop at ICML 2026 (poster). paper code
- Kalemati, M., **Noroozi, A.**, Shahbakhsh, A. et al. ParaAntiProt provides paratope prediction using antibody and protein language models. Sci Rep 14, 29141 (2024). link
- J. Reyes, **A. Noroozi**, Y. Xiao and M. Kersten-Oertel, Bridging the Gaps: Imputation of Parkinson's Disease Clinical Assessments With Federated Learning, in IEEE Journal of Biomedical and Health Informatics. link

Working Experience

310.ai | *AI Engineer in Computational Biology* September 2024 – October 2025
- Working on Protein Design Deep learning models
- ML research on Taxonomy prediction models.
- Designing Protein Binders using AI
- Developing AI chatbot backend using LLM calling Tools

Internship@Concordia | *Deep Learning, Medical Imaging* June 2023 – September 2023
- Summer Internship in Machine Learning and medical data, Medical imaging

Ershan Total Solution | *PHP, Back-end, Web developer* June 2022 – September 2022
- Developing website for trade in stocks using Laminas Framework in Php.
- my Last project was Excel and Pdf generator for all reports.

AI Medical | *Deep Learning, Medical Imaging* June 2021 – September 2021
- Summer Internship in Machine Learning and medical data, Medical imaging

Salam High school | *Mathematics Olympiad tutor* September 2021 – March 2023
- Teaching Mathematical Olympiad in high schools (Geometry, Algebra, combinatorics, Number Theory).

Honors

- Awarded the 2025 Presidential Fellowship at Koç University for MSc Entrance.
- Awarded the 2024 Vector Scholarship for MSc Entrance at York University, valued at 17,500 CAD.
- Acquired Bronze (2018) and Silver (2019) medals in **Iran's national Mathematical Olympiad**.
- Ranked 31 among 150k+ participants in Iran's national university entrance in 2020 (AKA. KONKUR).

Reviewing

- Reviewer, GenBio Workshop (Generative and Agentic AI for Biology), ICML 2026
- Reviewer, ML4SM Workshop, NeurIPS 2026

Research Experience

DELTA Team, Inria Nancy | *Protein Binder Design, Molecular Dynamics* June 2026 – Present

- Summer research internship under supervision of Hamed Khakzad and Yasaman Karami
- Developed **IDPepDesign**, a target-conditioned shortening diffusion model for peptide binder design (submitted to NeurIPS 2026)
- Evaluating protein binder design candidates using **molecular dynamics simulation** to assess protein flexibility and binding quality
- Contributing to computational biology research on protein design and binder assessment pipelines

AMP-DiT | *Generative Modeling, Diffusion Transformers, Computational Biology* October 2025 – Present

- Koç University, Computer Engineering; supervised by Prof. Attila Gürsoy and Prof. Özlem Keskin
- Developed **AMP-DiT**, a denoising diffusion transformer for antimicrobial peptide generation directly in sequence space
- Guided generation with an AMP classifier to produce diverse peptides with strong predicted antimicrobial activity
- Accepted as a poster at the GenBio Workshop at ICML 2026

Taxonomy Lineage Generation | *Transformer Models, Protein Language Models, Sequence-to-Sequence Learning* 2024

- AI Research Engineer, 310.ai
- Built a **transformer-based model** that generates full **taxonomy lineages** from protein sequences.
- Designed training pipelines, tokenization strategies, and loss functions to stabilize sequence generation.
- Addressed token collapse issues (model repeating most frequent token) and improved generalization via curriculum learning and embedding regularization.

TED Domain Classifier | *Protein Domain Classification, Fine-Tuning, Multiclass Modeling* 2024

- AI Research Engineer, 310.ai
- Project Link
- Created a **multi-class classifier** for TED immunoglobulin domains using protein LMs (ProtT5, ESM-2).
- Ran large-scale data preprocessing, domain extraction, and fine-tuning experiments on GPU clusters.

IgG1 Binder Design | *Protein Design, Generative AI, Structural Modeling* 2024

- AI Research Engineer, 310.ai
- Designed **IgG1 antibody binders** using generative protein models and sequence design pipelines.
- Integrated structural constraints and PLM embeddings to guide binder specificity and stability.

OPL | *Transfer Learning, Pytorch, Antibody, Paratope Prediction* June 2023 – September 2024

- Department of Computer Engineering, Sharif University
- Research as Bachelor Project under Supervision of Dr. Koohi
- ParaAntiProt: A Transfer Learning for Paratope Prediction only using Sequence
- Improving the accuracy of Paratope Prediction using Protein Language Models

apLab | *Federated Learning, Ml Medical, Pytorch, Parkinson Diseases* June 2023 – September 2023

- Department of Computer Science, Concordia University, Montreal City, Canada
- Research as Intern under Supervision of Dr.Kersten-Ortel
- Comparing and implementing different Federated Learning Algorithms for data imputation on Parkinson's Diseases.
- Using GAN and Machine Learning methods for Parkinson's disease prediction with distributed data

apLab | *Federated Learning, Distributed Deep Learning, Pytorch* June 2023 – September 2023

- Department of Computer Science, Concordia University, Montreal City, Canada
- Research as Intern under Supervision of Dr.Kersten-Ortel
- Implementing PW, FedAvg, FedOpt, ..., in Pytorch and working on Mathematically proof of PW Convergence
- Developing federated Algorithms for increasing performance in non-identical data

OPL | *Attention Deep Learning Models, Protein Affinity Binding, Pytorch, MHC-Peptide* Dec 2022 – Mar 2023

- Department of Computer Engineering, Sharif University
- Research as a Bachelor Student under the Supervision of Dr.Koohi

Language Proficiency

- Persian: Native
- English: Fluent, TOEFL (T:97, R:26, L:25, S:25, W:21)

Teaching Assistant Experience

- Teaching Assistant Python Programming, Koç University
- Head of Teaching Assistant Team of Probability and Statistics for 2 Semester (lecturer: Dr.Sharifi-Zarchi)
- Head of Teaching Assistant Team of Compiler Design (lecturer: Dr.Hosseinmardi)
- Project's Head in Teaching Assistant Team of Intro to Bioinformatics
- Discrete Structures, Probability and Statistics (2 times), Basic programming Python